



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to TIRUCHENGODU AGRO PDT CO- OP MKTG SO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

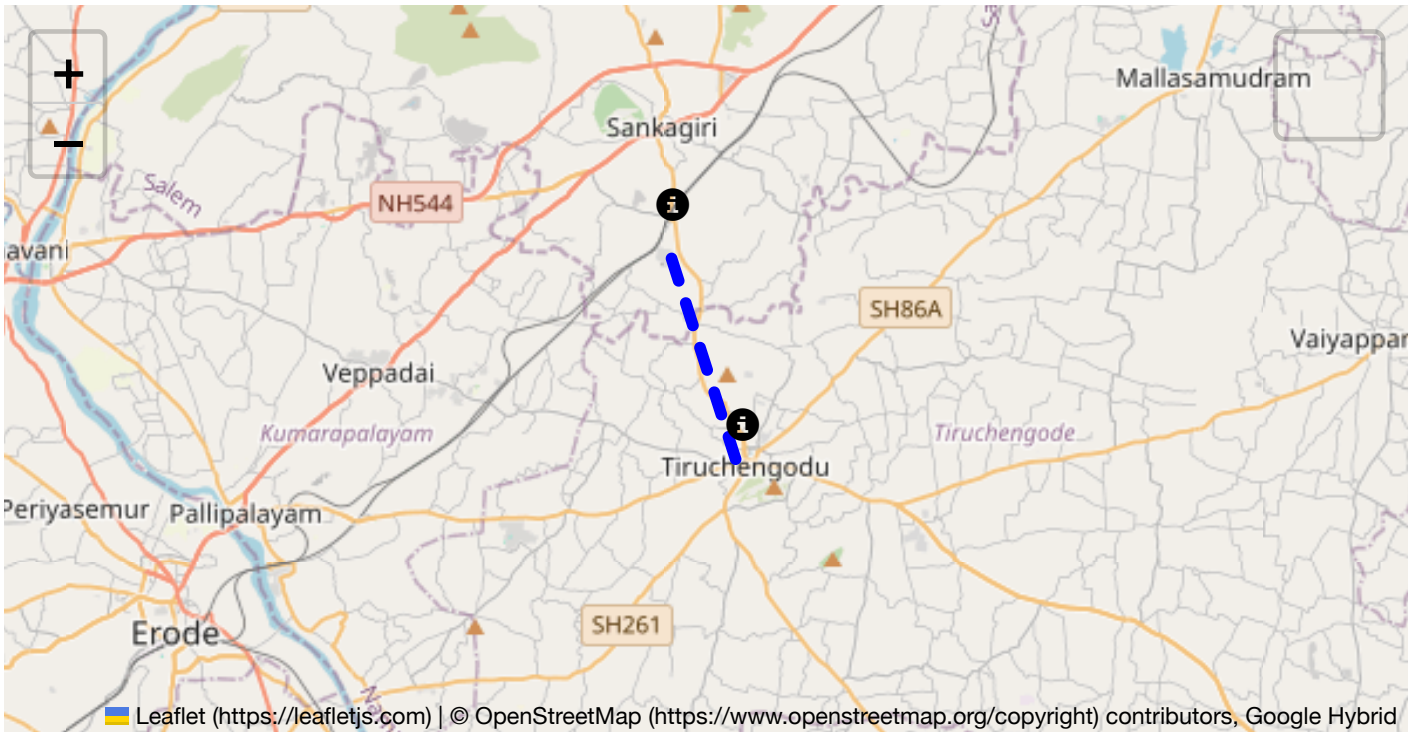
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

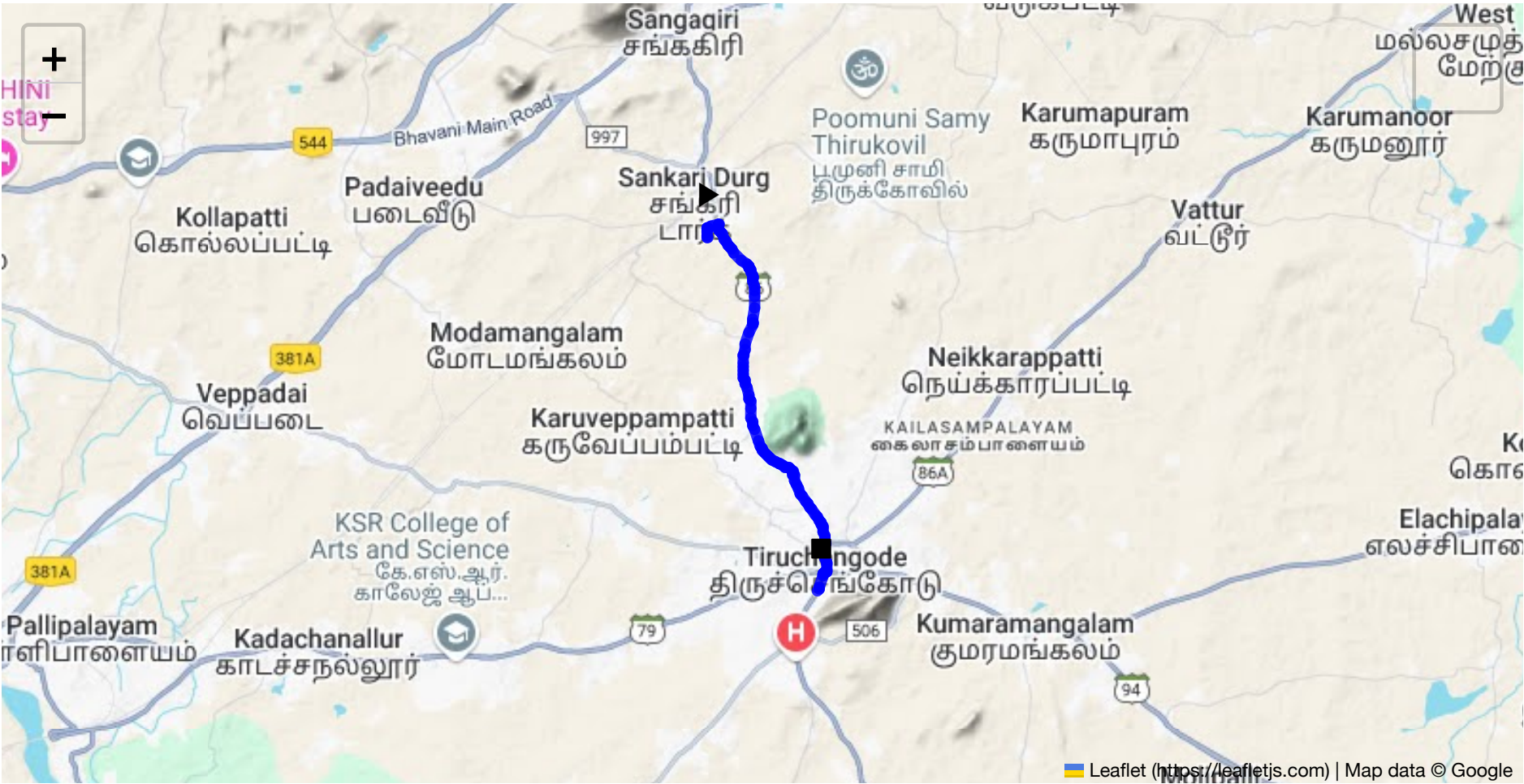


Route Summary:
Total Distance: 8.50 km
Estimated Duration: 0.3 hours

Adjusted Duration (Heavy Vehicle): 0.4 hours

Start: (11.4381, 77.8734)

End: (11.37526, 77.893732)



Welcome to the Journey Risk Management Study

To provide a detailed safety analysis of the route from CVQF+23W, Sangagiri to 9VGV+4G2, Kamalar, Tiruchengode in Tamil Nadu, we will explore each point in your request.

- Overview of the Route Map:** The route spans approximately 8.50 kilometers, primarily along regional roads with local traffic. It involves driving through rural and semi-urban areas of Tamil Nadu, with typical waypoints and landmarks along the route.
- Typical Weather Conditions and Potential Weather-related Hazards:** Tamil Nadu experiences a tropical climate, with hot and humid summers, moderate monsoons, and mild winters. During monsoon months (June to September and October to November), heavy rains can lead to slippery roads and localized flooding. It's crucial to watch for weather updates before and during the drive.
- Analysis of Traffic Patterns:** The route does not usually experience heavy traffic congestion compared to urban centers, but local peak hours (morning around 8-10 AM and evening 5-7 PM) might slow down progress due to school and work commutes. Small towns may have bottlenecks, especially at markets or junctions.
- Assessment of Road Quality and Infrastructure:** The road quality along the route is generally variable. While some segments have been maintained well, others may have potholes or narrow passages that require cautious driving. Signage might be limited, and road lighting can be sparse outside of populated areas.
- Suggestions for Alternative Routes for Emergencies:** In case of roadblocks or emergencies, alternative routes could branch through local villages linking back to main roads; however, these might be less developed. It's advisable to have a local GPS or navigation aid to navigate these alternatives.
- Summary of Local Regulations Affecting Hazardous Material Transport:** Tamil Nadu, like other Indian states, has strict regulations on transporting hazardous materials, including permits, specific

routes, and timing restrictions to avoid populated areas during peak hours. Ensure all paperwork and regulatory requirements are in order.

- 7. **Overview of Historical Incidents:** There have been occasional reports of vehicle accidents due to road conditions and driver error in the region. The lack of data on incidents involving heavy or hazardous vehicles specifically along this route emphasizes the need for driver vigilance.
- 8. **Environmental Considerations and Sensitive Areas:** There are agricultural zones along the route, so it's vital to prevent any spillage or contamination. Bathing and grazing areas need to be avoided due to their environmental sensitivity.
- 9. **Analysis of Communication Coverage:** The route generally has decent cellular coverage, but some areas, especially rural segments, might have weaker signals. It's advisable to have a reliable mobile network carrier and in-vehicle communication devices.
- 10. **Estimated Emergency Response Times:** Emergency response might vary between rural and semi-urban sections, typically taking 20-40 minutes given proximity to Tiruchengode or other nearby towns. It's advisable to identify hospitals and police stations along the way.
- 11. **Overall Summary of Risk Assessment:** The risk along this route is moderate, primarily due to road quality and weather conditions during certain seasons. Good planning, adherence to regulations, cautious driving, and awareness of alternative routes can manage risks effectively. It's crucial for the truck driver to stay informed about current road conditions and weather forecasts, maintain communication with logistics teams, and keep emergency contacts readily accessible.

Overall, with careful planning and adherence to safety and regulatory measures, the journey can be navigated safely.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453

